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# Hydrolysis of cellulose from sugarcane bagasse by cellulases from marine-derived fungi strains

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A B S T R A C T

Four marine-derived fungi strains—*Penicillium citrinum* CBMAI 1186, *Aspergillus sydowii* CBMAI 934, *Aspergillus sydowii* CBMAI 935 and *Mucor racemosus* CBMAI 847—were studied as potential microbial producers of cellulases under solid-state fermentation. The cultivation parameters (time, pH and temperature) and the performance of the cellulases in the hydrolysis of cellulose from sugarcane bagasse were evaluated. The best cultivation time in solid-state fermentation for producing cellulases was 3 days for all of the strains except for *A. sydowii* CBMAI 934, which exhibited its highest filter paper activity (FPU) after 7 days. After optimizing pH and temperature, activities of 0.73, 1.9, 2.4 and 1.3 FPU $g^{-1}$ (of substrate) were obtained for *A. sydowii* CBMAI 935, *A. sydowii* CBMAI 934, *M. racemosus* CBMAI 847 and *P. citrinum* CBMAI 1186, respectively. The hydrolysis of cellulose was studied using rind and pith fractions of sugarcane bagasse, with and without alkali pretreatment. Independent of the fungi strains, untreated bagasse was resistant to saccharification. After the treatment, the degree of saccharification was 78% for *A. sydowii* CBMAI 934 and 24% for *M. racemosus* CBMAI 847. The different anatomical and morphological characteristics of the rind and pith cells influenced the saccharification by cellulases for the different marine fungi strains. The hydrolysis by cellulases from *A. sydowii* CBMAI 934 produced 56% and 81% saccharification in the rind and pith, respectively. This fungus strain exhibits significant potential for the production of fermentable sugars resulting from the hydrolysis of cellulose from sugarcane bagasse, mainly after pretreatment with NaOH solution.



## 1. Introduction

Cellulases are enzymatic complexes composed of endoglucanases, exoglucanases and β-glucosidases. They are mainly produced by microorganisms such as fungi and bacteria and they act synergistically to cleavage β-1,4-glycosidic bonds in cellulose chains under mild conditions and with high specificity (Binod et al., 2011; Saini et al., 2015). Hong et al. (2015) evaluated 50 fungi species from the marine environment and found that *Arthrinium saccharicola* exhibited a Filter Paper Unity (FPU) cellulase activity of 0.39 U $mL^{-1}$. Other marine-derived fungi species reported to be cellulase producers include *Aspergillus niger* (Raghukumar et al., 2004; Xue et al., 2012, 2015; Sethi et al., 2013), *Aureobasidium pullulans* (Chi et al., 2009), *Aspergillus terreus*, *Mucor plumbeus* (Padmavathi et al., 2012), *Chaetomium globosum* (Ravindran et al., 2011), *Cladosporium sphaerospermum* (Trivedi et al., 2015) and *Alternaria alternata* (Faten and El Aty Abeer, 2013).

The use of cellulases in the hydrolysis of a wide range of lignocellulosic raw materials (e.g., corn stover, rice straw, sweet sorghum bagasse) to obtain fermentable sugar has been the subject

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of several studies (e.g., Chen et al., 2009; Sukumaram et al., 2009; Laothanachareon et al., 2015; Sun et al., 2015). Although many efforts have focused on the development of enzymatic systems to perform the hydrolysis of lignocellulosic raw materials, some technical limitations in the conversion of biomass to bioethanol persist (Saini et al., 2015).

Economical issues are also one of the challenges in bioethanol production from biomass and the use of cheaper raw materials for microorganism cultivation, such as those cited above, and the use of cost-effective fermentation strategies like solid-state fermentation (SSF), which enables higher cellulase production compared with submerged fermentation (SmF), have great potential to overcome this challenge (Sukumaram et al., 2009; Subramaniyam and Vimala, 2012; Jain et al., 2013).

Another important barrier to the implementation of the lignocellulosic raw materials in bioethanol production are their complex structures. The composite matrix of lignin, hemicelluloses and cellulose demands the development of pretreatment strategies to make the cellulose more accessible for enzymatic hydrolysis, as observed by Chen et al. (2009) in the treatment of corn stover. One of the methods described in the literature for the pretreatment of lignocellulosic raw materials is the use of a diluted sodium hydroxide (NaOH) solution. This method has been shown to be very efficient for lignin removal and produces less inhibitors for the fermentation step than using an acidic treatment. A diluted NaOH solution pretreatment is also associated with an increase in sugar production during enzymatic hydrolysis for different substrates (Chen et al., 2009; Sukumaram et al., 2009).

In this work, four marine-derived fungi strains, *Penicillium citrinum* CBMAI 1186, *Aspergillus sydowii* CBMAI 935, *Aspergillus sydowii* CBMAI 934 and *Mucor racemosus* CBMAI 847, were studied for the production of cellulases under SSF. The enzymatic hydrolysis of the pretreated sugarcane bagasse and the release of reducing sugars were also evaluated. To the best of our knowledge, these four marine-derived fungi strains have not yet been evaluated as potential cellulase producers.

## 2. Materials and methods

### 2.1. Fungi strains and inoculum preparation

The marine-derived fungi *A. sydowii* CBMAI 934 and *A. sydowii* CBMAI 935 were isolated from the marine sponge *Chelonaplysylla erecta*; *P. citrinum* CBMAI 1186 was isolated from the marine alga *Caulerpa* sp., and *M. racemosus* CBMAI 847 was isolated from Brazilian *Cnidarian* specie *Mussismilia hispida*. All fungi were collected in São Sebastião, São Paulo-Brazil, by Prof. Roberto G. S. Berlinck (IQSC-USP). For the inoculums, it was used a spore solution, whose spores were counted using direct microscopy using a Neubauer cell counter chamber.

### 2.2. Solid-state fermentation (SSF)

For preparing the SSF medium, 5 g of commercial wheat bran and 10 mL of Mandel solution (pH 6.0) were added to 125-mL Erlenmeyer flasks. The Mandel solution (Mandels and Weber, 1969) was composed of peptone (0.75 g $L^{-1}$), urea (0.30 g $L^{-1}$), $(NH_4)_2SO_4$ (1.40 g $L^{-1}$), $KH_2PO_4$ (2.00 g $L^{-1}$), $MgSO_4.7H_2O$ (0.30 g $L^{-1}$), $CaCl_2.2H_2O$ (0.40 g $L^{-1}$), $ZnSO_4$ (1.40 mg $L^{-1}$), $FeSO_2.7H_20$ (5.00 mg $L^{-1}$), $CoCl_2.6H_20$ (2.00 mg $L^{-1}$) and $MnSO_4.5H_20$ (1.60 mg $L^{-1}$). The culture medium was autoclaved (20 min, at 121 °C) and the strains were inoculated by adding a spore solution ($5.10^5$ spores/g wheat bran) to the culture medium. The fungi were cultivated for 2 days at 32 °C. All the experiments were performed in triplicate.

### 2.3. Submerged fermentation (SmF)

The SmF medium was prepared by adding 0.5 g of wheat bran to 125-mL Erlenmeyer flasks containing 50 mL of Mandel solution (pH 6). The culture medium was autoclaved (20 min, at 121 °C) and the flasks were inoculated with $10^5$ spores/mL and cultivated in an orbital shaker for 2 days at 32 °C and 130 rpm.

### 2.4. Enzyme extraction

Cellulase extraction from SSF was performed by adding 50 mL of acetate buffer (50 mM, pH 4.5) followed by magnetic stirring for 1 h (Dave et al., 2013). Then, the solid was filtered under low pressure and the solids remaining in the filtrate (crude extract) were removed by centrifugation (10,000 rpm for 6 min). For cellulase extraction from SmF, the liquid medium was stirred for 1 h and then filtered and centrifuged as well (10,000 rpm, 6 min).

### 2.5. Enzyme activity measurements

FPase (Filter Paper Activity) assay was performed via standard filter paper method described by Ghose (1987) and one unit was defined as the amount of enzyme required to release one μmol of glucose per minute after 1 h of enzymatic hydrolysis. The concentration of the released reducing sugars was determined using the dinitrosalicylic acid (DNS) method (Miller, 1959). The FPase activities were expressed in FPU $g^{-1}$ (solid media) or FPU $mL^{-1}$ (liquid media). The total volume used in the FPase assay in all experiments was 3 mL (i.e., 1 mL of crude extract and 2 mL of buffer solution). The standard values used in the Ghose method were pH 4.8 (citrate buffer, 50 mM), 50 °C and approximately 50 mg of filter paper (Whatman No.1, strips 6 × 1 cm).

Endoglucanase (CMCase) (using carboxymethylcellulose purchased from Sigma-Aldrich as substrate) and β-glucosidase (using cellobiose purchased from Sigma-Aldrich as substrate) activity were measured as described previously by Ghose (1987); the concentration of the released reducing sugars was also determined via the DNS method. The CMCase and β-glucosidase activity unit was defined as the amount of enzyme required to release one μmol of glucose per minute for the hydrolysis reaction carried out for 0.5 h.

### 2.6. Improved cellulase production in SSF

The influence of pH, temperature, volume of Mandel solution and the cultivation time in the cellulase production during the growth of the marine-derived fungi strains was evaluated. The studied pH range was 3–8, the temperature range was 25–45 °C (intervals of 5 °C; no growth was observed at 45 °C), the range of volume of Mandel solution was 7.5–17.5 mL (intervals of 2.5 mL) and cultivation time studied was 1–7 days. The experiments were performed in triplicate.

### 2.7. Optimization of enzymatic hydrolysis conditions on the FPase assay

Studies of improvement of the hydrolysis of the filter paper by the crude extracts produced by different marine-derived fungi strains were evaluated by changing the pH value (2.8, 3.8, 4.8, 5.8, 6.8 and 7.8) and the temperature (30, 40, 50, 60 and 70 °C) used during the hydrolysis of filter paper. Three buffer solutions were tested during the study of the pH influence in the hydrolysis of filter paper — citric acid/$Na_2HPO_4$ (50 mM) (pH 2.8 and 3.8), citrate buffer (50 mM) (pH 3.8, 4.8 and 5.8) and phosphate buffer (50 mM) (pH 5.8, 6.8 and 7.8). The influence of enzyme concentration in the hydrolysis of filter paper (50 mg) was also evaluated, by changing the amount of crude extract solution (0.5, 1.0, 1.5 and 2.0 mL). All the experiments were performed in triplicate.

### 2.8. Pretreatment of sugarcane bagasse for lignin removal

The integral sugarcane bagasse and its fractions (pith and rind) were pretreated with sodium hydroxide solution (2.06% NaOH (w/w)) in a ratio solid-liquid 1:10, using anthraquinone as catalyst (0.15% wt), at 160 °C for 100 min using stainless steel reactors.

To determine the chemical composition of the untreated and treated integral sugarcane bagasse and its fractions were submitted to the methodology described by Pasquini et al. (2005). In summary, the lignocellulosic materials were hydrolyzed by sulfuric acid (72%), generating a solid residue (Klason lignin) and a solution containing the sugars released from the polysaccharides. The lignin percentage in the samples was determined accordingly Pasquini et al. (2005) and the percentage of cellulose and hemicellulose were determined by analyzing the sugars content using a high performance liquid chromatograph (Shimadzu CR7A equipped with Aminex HPX-87H (Bio-Rad) column, RI detector (Shimadzu RID20A) and using $H_2SO_4$ 5 mM as eluent, 0,6 mL min$^{-1}$, at 45 °C).

### 2.9. Saccharification of sugarcane bagasse

The untreated and treated integral bagasse (as generated in sugarcane plant) and its fractions were tested and compared to evaluate the effect of the previous lignin removal on the saccharification of sugarcane bagasse. The saccharification was performed by using the optimum pH and temperature – determined during the filter paper hydrolysis – of each complex of cellulases produced by the marine-derived fungi strains. It was added the same FPase value (10 FPU g$^{-1}$ of substrate) for all the extract of cellulases. The experiments were performed using a total volume of 50 mL (crude extract + buffer) containing 100 mg of the lignocellulosic materials (dry mass). After 72 h, the amount of released reducing sugar was determined via DNS method using glucose as equivalent, and the remaining material was dried and weighed to determine the percentage of saccharification.

### 2.10. Statistical analysis

The analysis of variance (ANOVA) was performed by using Microsoft Office Excel 2013 and Tukey test was performed by OriginPro 7.0 software.

## 3. Results

### 3.1. Marine-derived fungi strains as cellulase producers under SSF and SmF

The marine-derived fungi strains *A. sydowii* CBMAI 934, *A. sydowii* CBMAI 935, *P. citrinum* CBMAI 1186 and *M. racemosus* CBMAI 847 exhibited different FPase values when cultivated under SSF or SmF conditions (Table 1). Subramaniyam and Vimala (2012) stated that cultivation by SSF produces more enzymes than cultivation by SmF; this assertion was confirmed in this study (Table 1). SmF cultivation was associated with lower FPU activity values for all of the fungi strains compared with those obtained by SSF cultivation (3-fold for *A. sydowii* CBMAI 934 and *A. sydowii* CBMAI 935; 4-fold for *P. citrinum* CBMAI 1186; 2-fold for *M. racemosus* CBMAI 847). *P. citrinum* CBMAI 1186 exhibited the highest FPase value (1.20 FPU g$^{-1}$) of the studied marine-derived fungi strains under SSF cultivation. Given that the highest values of cellulase activity were observed for SSF cultivation, all additional experiments were performed using this type of cultivation.

### 3.2. The influence of cultivation conditions under SSF on cellulase production

#### 3.2.1. Cultivation time

Once the ability of the marine-derived fungi strains to hydrolyze cellulose from filter papers was determined, the best cultivation conditions for each fungus as a cellulase producer were investigated. The optimum cultivation period for *A. sydowii* CBMAI 935, *P. citrinum* CBMAI 1186 and *M. racemosus* CBMAI 847 was 3 days, and the optimum cultivation period for *A. sydowii* CBMAI 934 was 7 days (Fig. 1).

The FPase of *A. sydowii* CBMAI 934 increased, even considering the experimental error, over the cultivation period of 7 days (Fig. 1a). This contrasted with the situation observed for the other three fungi strains studied, which exhibited maximum FPase after 3 days (Fig. 1b–d). For this reason, the cultivation time for all of the marine-derived fungi strains was selected to be 3 days, even for *A. sydowii* CBMAI 934. On the third day of cultivation, this strain exhibited already about 70% of the maximum activity that was observed after 7 days of cultivation. In these experiments, at the 0.05 level, the population means were significantly different in the ANOVA.

#### 3.2.2. Influence of initial pH value

Modulating the initial pH value of the Mandel solution between 3 and 8 did not result in a regular pattern of the relative FPase for the different fungi strains that were tested (Fig. 2). The marine-derived fungus *A. sydowii* CBMAI 934 exhibited better activity for acidic pH values (3 and 4), which were statistically different of the results obtained in the pH range between 5 and 8 (Fig. 2a), accordingly the Tukey test. *A. sydowii* CBMAI 935 demonstrated almost the same activity values over the entire pH range (Fig. 2b) once that only the cultivation in pH 4 and pH 8 were statistically different from the other values (Tukey test). *P. citrinum* CBMAI 1186 exhibited good relative FPase for acidic pH values (pH 3 and 4), but its highest activity was observed at pH 8 (Fig. 2c). *M. racemosus* CBMAI 847 exhibited irregular variation, with better activity at pH 4 and 6, and lower levels of activity at pH values of 3, 5, 7 and 8 (Fig. 2d). At the 0.05 level, the population means were significantly different for all the experiments.

#### 3.2.3. Influence of cultivation temperature

*A. sydowii* CBMAI 934 exhibited its best activity when cultivation was carried out at 30 °C (Fig. 3a). The best cultivation temperature for cellulase production by *A. sydowii* CBMAI 935 was 40 °C; this strain also exhibited good activity at 25 °C and 30 °C (Fig. 3b). *P. citrinum* CBMAI 1186 exhibited good cellulase production at 25 °C and 35 °C; its highest relative activity was measured at 25 °C (Fig. 3c). *M. racemosus* CBMAI 847 also exhibited good relative

**Table 1**
FPase values of the crude extract obtained from cultivation of the fungi strains under SSF and SmF using wheat bran as carbon source.

| Strain | SSF (FPU g$^{-1}$) | SSF[a] (FPU mL$^{-1}$) | SmF[b] (FPU mL$^{-1}$) |
|---|---|---|---|
| *A. sydowii* CBMAI 934 | 0.80 ± 0.07 | 0.080 ± 0.007 | 0.025 ± 0.001 |
| *A. sydowii* CBMAI 935 | 0.60 ± 0.14 | 0.060 ± 0.014 | 0.019 ± 0.001 |
| *P. citrinum* CBMAI 1186 | 1.20 ± 0.11 | 0.120 ± 0.011 | 0.029 ± 0.001 |
| *M. racemosus* CBMAI 847 | 0.60 ± 0.04 | 0.060 ± 0.004 | 0.030 ± 0.003 |

[a] Total volume of 50 mL of acetate buffer used for enzyme extraction.
[b] Total volume of liquid of 50 mL obtained after filtration.

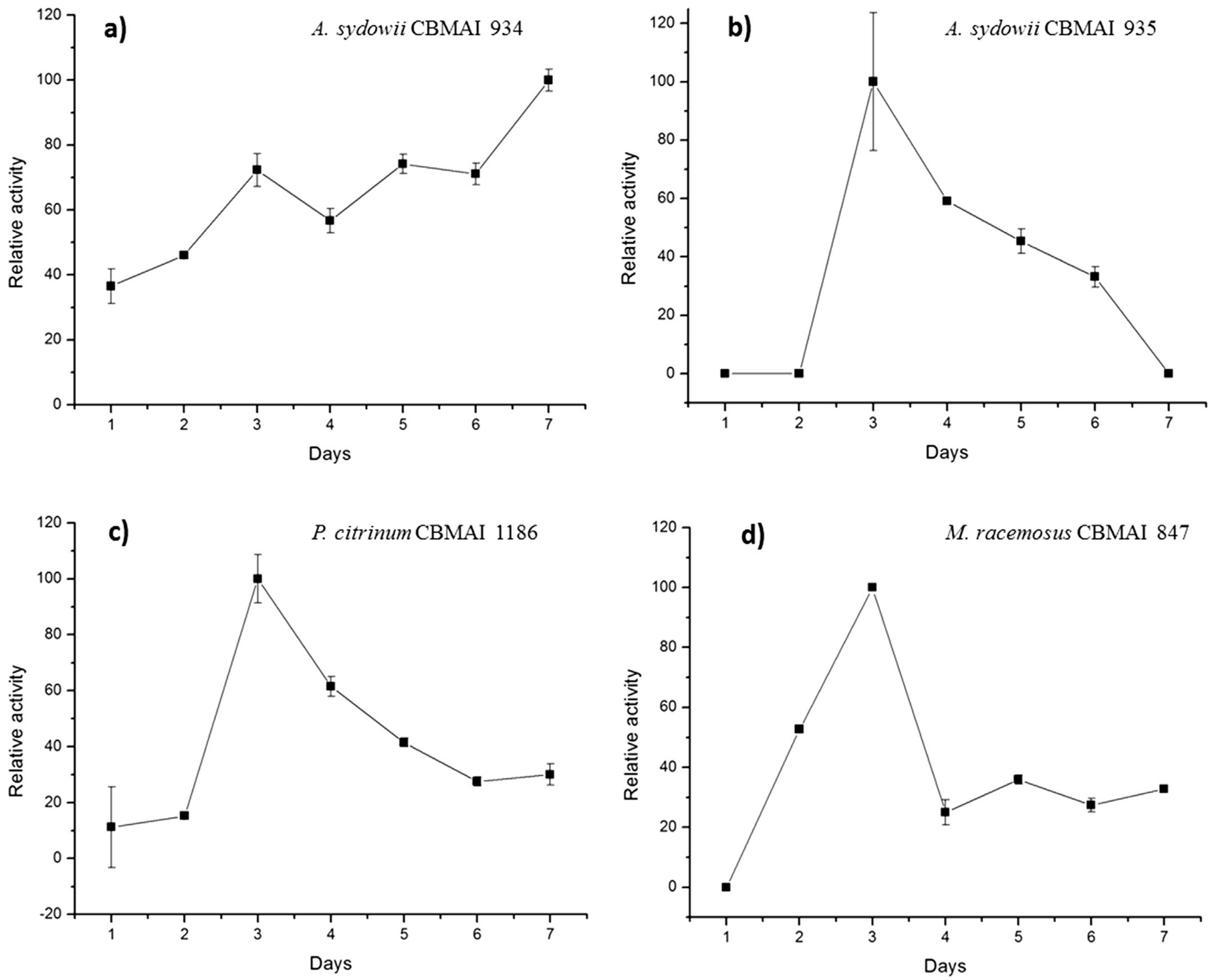


**Fig. 1.** Relative activity of crude extract obtained with different fungi strains along of cultivation period in SSF[a]. [a]The activity 100% was the maximum FPase obtained along of cultivation period. The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (186,2), *A. sydowii* CBMAI 935 (102,7), *P. citrinum* CBMAI 1186 (127,8) and *M. racemosus* (1642,1).

activity at 25 °C and 35 °C, but its highest FPase was obtained at 35 °C (Fig. 3d). Overall, the best temperature was therefore found to be in the range of 25–40 °C, which can be explained by the fact that these fungi strains are mesophile microorganisms, with better growth between 20 °C and 40 °C (Hung et al., 2005). In these experiments, at the 0.05 level, the population means were significantly different.

#### 3.2.4. Influence of volume of Mandel solution

In previous SSF experiments, the volume of Mandel solution used was 10 mL per 5 g of wheat bran. The best relative activity for *A. sydowii* CBMAI 934 and *M. racemosus* CBMAI 847 was obtained when 7.5 mL and 12.5 mL of Mandel solution was used, respectively (Fig. 4a and d). In contrast, *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186 exhibited good adaptability and cellulase production for all volumes of Mandel solution tested in this study (Fig. 4b and c). In these experiments, at the 0.05 level, the population means were significantly different.

### 3.3. The influence of enzymatic hydrolysis conditions on the FPase assay

#### 3.3.1. Influence of pH value

In all previous experiments, the FPase assay was performed according to the methodology of Ghose (1987); (citrate buffer 50 mM, pH 4.8, 50 °C). For *A. sydowii* CBMAI 934, *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186, the best relative FPase were obtained using citrate buffer, with pH values of 3.8, 4.8 and 5.8, respectively (Fig. 5a–c). For *M. racemosus* CBMAI 847, two maximum FPU activities were observed at pH 4.8 with citrate buffer and at pH 7.8 with phosphate buffer (Fig. 5d). In these experiments, at the 0.05 level, the population means were significantly different.

#### 3.3.2. Influence of temperature

*A. sydowii* CBMAI 934 exhibited its best relative activity at 40 °C (Fig. 6a), while *A. sydowii* CBMAI 935 presented similar values of FPU activity between 30 °C and 70 °C (Fig. 6b). *P. citrinum* CBMAI 1186 exhibited higher activity between 50 °C and 60 °C (Fig. 6c).

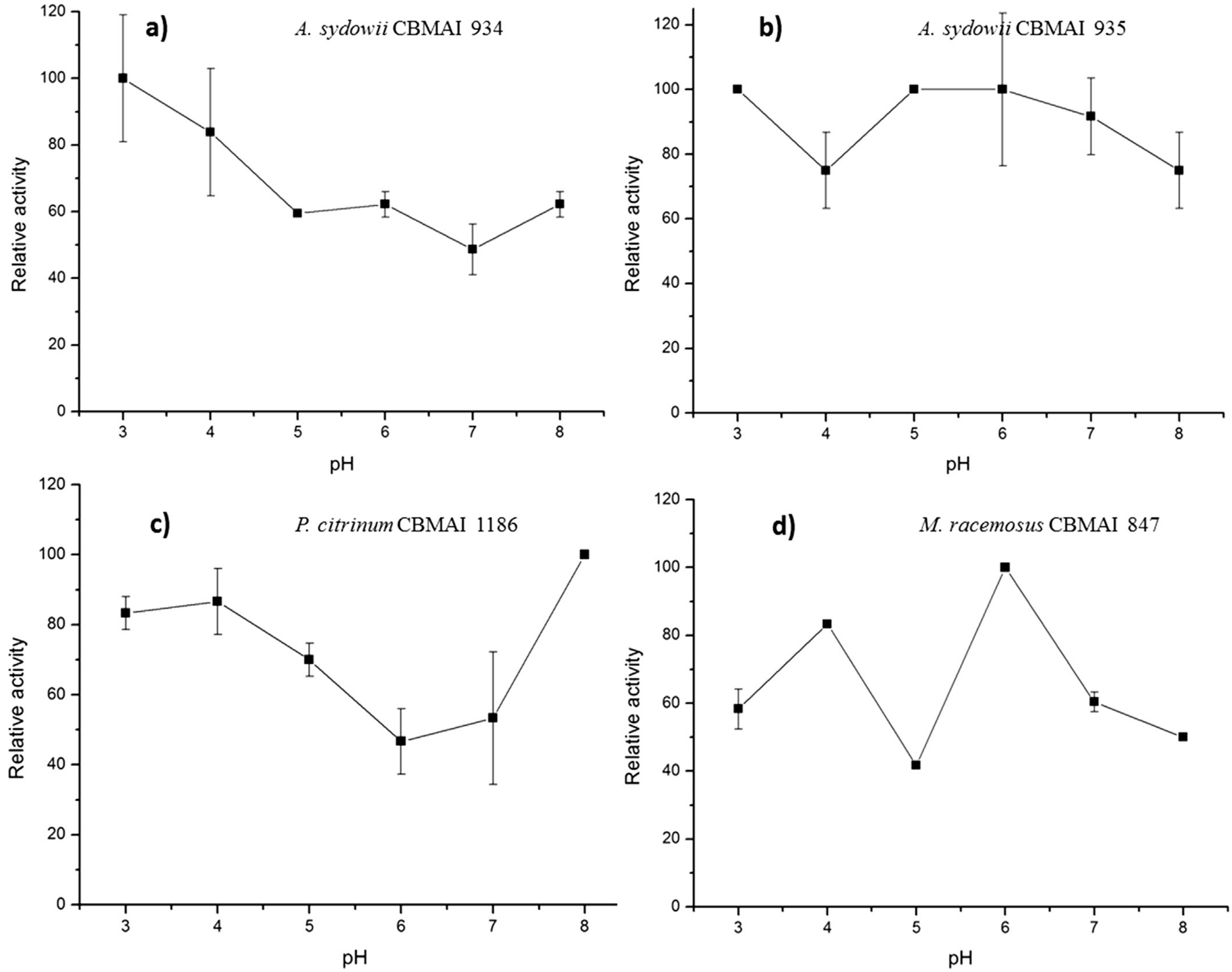


**Fig. 2.** Relative activity of crude extract obtained with different fungi strains by changing initial pH value of Mandel solution in SSF[a]. [a]The activity 100% was the maximum FPase obtained for each pH value tested. The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (15,1), *A. sydowii* CBMAI 935 (5,6), *P. citrinum* CBMAI 1186 (26,3) and *M. racemosus* (387,1).

*M. racemosus* CBMAI 847 demonstrated its best relative activity at 30 °C (Fig. 6d). In these experiments, at the 0.05 level, the population means were significantly different, except for the experiments using cellulases produced by the strain *A. sydowii* CBMAI 935.

### 3.3.3. Influence of enzyme concentration

The influence of enzyme concentration was analyzed based on the volume of crude extract used in the FPU assay. The best relative activity for all of the marine-derived fungi strains was obtained when 2 mL of crude extract was used (i.e., double the initial volume). Crude extract volumes of 0.5, 1.0 and 1.5 mL did not result in a significant change in FPase value for enzyme extracts produced by *A. sydowii* CBMAI 934, *A. sydowii* CBMAI 935 or *P. citrinum* CBMAI 1186 (Fig. 7a–c). On the other hand, for *M. racemosus* CBMAI 847, the FPase enhances proportionally with the increase of the volume of crude extract (Fig. 7d). In these experiments, at the 0.05 level, the population means were significantly different.

## 3.4. Profile of FPU, CMCase and β-glucosidase activity

The FPU, CMCase and β-glucosidase activity in crude extracts obtained from the different fungi strains were assayed (Table 2). The highest, among all the fungi strains, FPU activity was obtained from enzymatic crude extract of *M. racemosus* CBMAI 847 (2.4 FPU $g^{-1}$), which, after optimization during this study, was four times greater than its initial FPU values (Table 1). The lowest FPU activity in enzymatic crude extract was obtained from *A. sydowii* CBMAI 935, which was initially 0.60 U.$g^{-1}$ and only increased to 0.73 FPU $g^{-1}$. The FPase for *A. sydowii* CBMAI 934 increased from 0.8 FPU $g^{-1}$ to 1.9 FPU $g^{-1}$ and crude extract from *P. citrinum* CBMAI 1186 increased from 1.2 FPU $g^{-1}$ to 1.3 FPU $g^{-1}$.

CMCase and β-glucosidase activity recorded for the crude extracts obtained from these fungi strains did not show the same pattern of FPase (Table 2). *A. sydowii* CBMAI 935 exhibited the lowest FPase but the highest β-glucosidase activity, and its CMCase activity was almost as good as that obtained for *A. sydowii* CBMAI 934 and *M. racemosus* CBMAI 847. Another important finding was

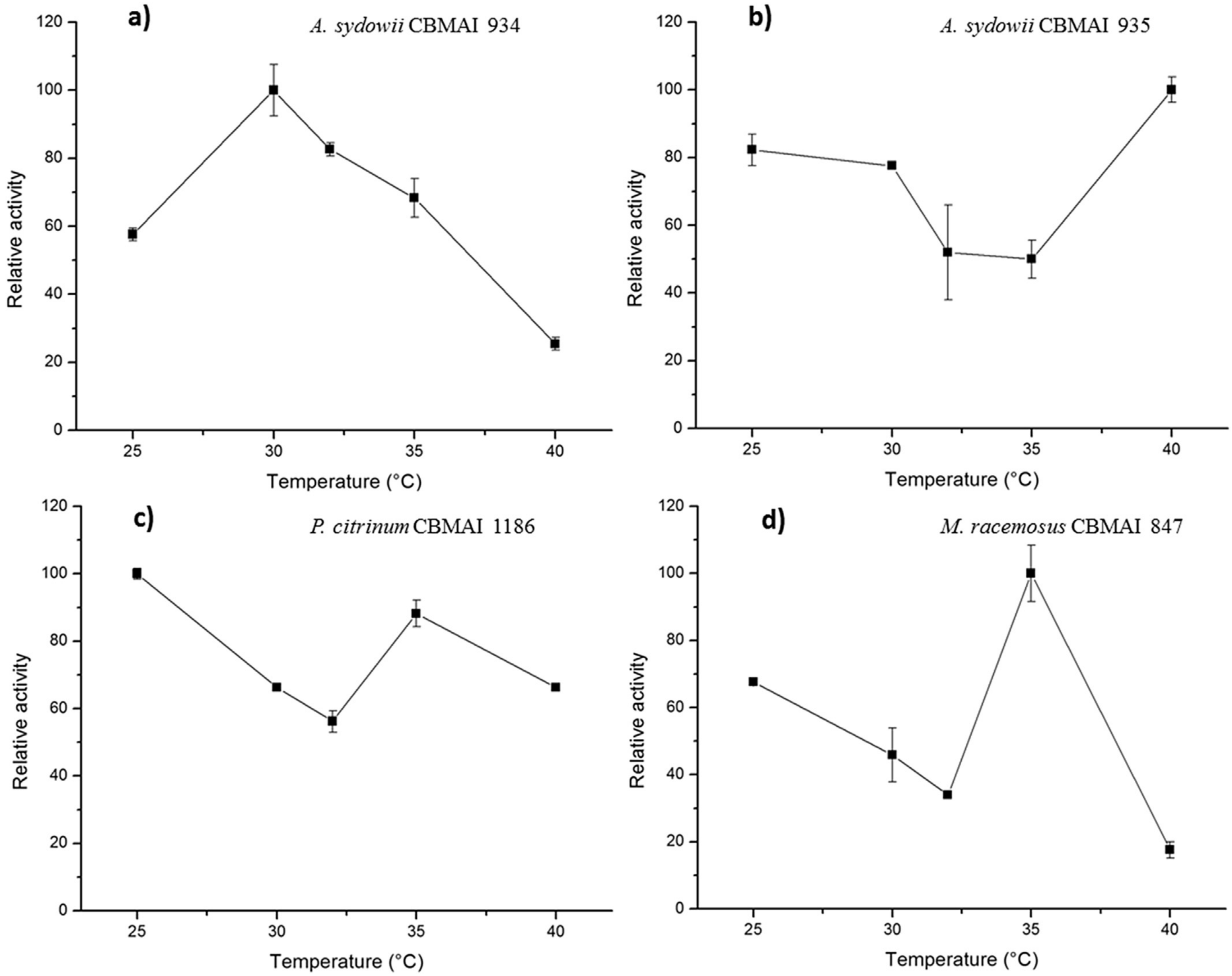


**Fig. 3.** Relative activity of crude extract obtained with different fungi strains by changing cultivation temperature in SSF[a]. [a]The activity 100% was the maximum FPase obtained for each value of cultivation temperature tested. The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (235,1), *A. sydowii* CBMAI 935 (51,8), *P. citrinum* CBMAI 1186 (344,4) and *M. racemosus* (217,8).

that the β-glucosidase activity for *A. sydowii* CBMAI 934 was very low when compared to its FPase and CMCase activities.

### *3.5. The influence of the second washing on the extraction of cellulases from solid nutrients*

To enhance the extraction of cellulases, a second step of washing of the solid substrate using the same volume of buffer (50 mL of acetate buffer, 0.05 M, pH 4.5) was carried. This second washing step showed that there was a significant amount of cellulases left adsorbed on wheat bran after the first washing (Table 3). After the second washing for *A. sydowii* CBMAI 934, 60% of remaining activity relative to initial activity was obtained. The strain that exhibited the smallest remaining activity was *M. racemosus* CBMAI 847 (25.1%). Since the recovery yield of cellulases is an important step after SSF, a second step of extraction should be implemented. However, it is necessary to take into account the economics of this additional extraction step.

### *3.6. Saccharification of sugarcane bagasse*

This is the first report on the application of marine-derived fungi on the saccharification of sugarcane bagasse and releasing of reducing sugars. The hydrolysis of cellulose present in the sugarcane bagasse was evaluated in three fractions that were: integral (untreated, the form in which it was collected, and pretreated), pith (pretreated) and rind (pretreated). The chemical composition and the degree of saccharification of raw and treated samples are displayed in Table 4.

Cellulases from *A. sydowii* CBMAI 934 and *M. racemosus* CBMAI 847 exhibited the best results in terms of hydrolyzing untreated bagasse; the percentage of saccharification was 20% (w/w) of dry mass. The cellulases produced by *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186 were unable to perform the same reaction. However, after pretreatment with sodium hydroxide/anthraquinone for 100 min, both of the crude extracts, from *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186, achieved 19% saccharification of the integral sugarcane bagasse, and 32% and 28% of the pith,

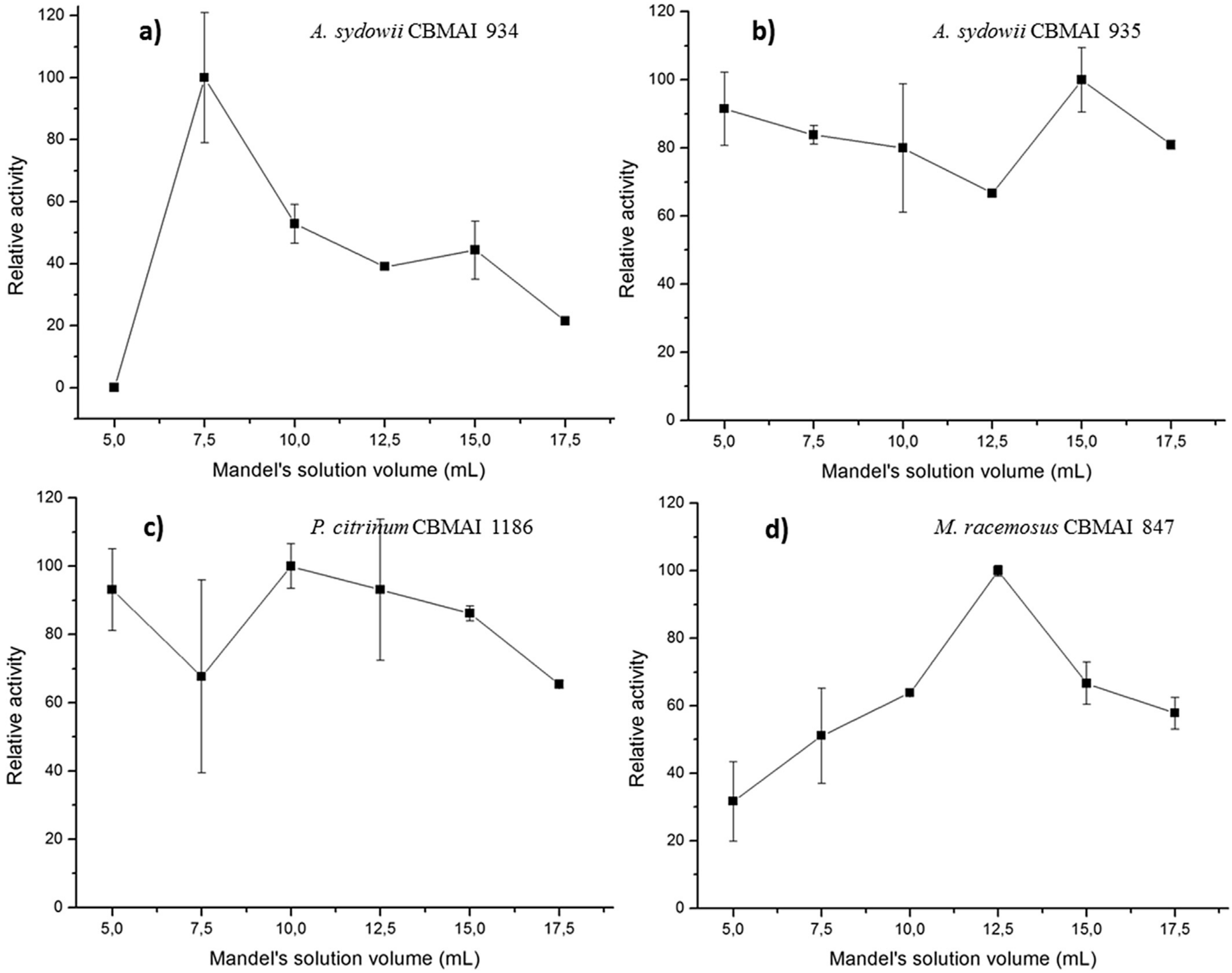


**Fig. 4.** Relative activity of crude extract obtained with different fungi strains by changing Mandel solution volume in SSF[a]. [a]The activity 100% was the maximum FPase obtained for each value of cultivation volume tested. The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (70,3), *A. sydowii* CBMAI 935 (8,0), *P. citrinum* CBMAI 1186 (5,3) and *M. racemosus* (45,0).

respectively. The cellulase from the fungus *A. sydowii* CBMAI 934 improved its hydrolytic activity after pretreatment, hydrolyzing 78%, 81% and 56% of the dry mass of the integral, pith and rind fractions from sugarcane bagasse, respectively. It is important to note that the pretreatment did not improve the efficiency of *M. racemosus* CBMAI 847; the saccharification degree only increased from 20% to 24%.

## 4. Discussion

The best results were obtained for a cultivation time of 3 days (Fig. 1); this incubation period is shorter than some periods found in the literature (i.e., 2–21 days; Dhillon et al., 2011; Dhillon et al. 2012; Faten and El Aty Abeer, 2013). Cultivation time has an important effect on cellulase production; even comparing a wild strain and its mutants, they may have different periods for biosynthesizing and secreting cellulases according to the type of solid nutrient used and the incubation conditions (Raghuwanshi et al., 2014). In the study reported by Raghuwanshi et al. (2014) it was found that the mutant fungus strain exhibited a higher FPase until the 5th day, after which it exhibited the same activity as the wild strain. For this reason, the best cultivation time for cellulase production was determined separately for each one of the marine-derived fungi strains accordingly their highest FPase value.

The effect of initial pH in the cultivation of cellulase production did not exhibit a consistent pattern (Fig. 2). The cellulases from *M. racemosus* CBMAI 847 presented two optimum pH values (Fig. 2d), and this behavior has not been reported in the literature until now. In the future studies, this finding should be explored and clarified whether it is specific only for this marine-derived fungus strain. The two optimal pH values obtained may be due to: first changes in the availability of carbon source in the media that demand other cellulase enzymes when the pH value is altered, or second due to the different amount of individual cellulases that changes the behavior and the characteristic activity profile of biosynthesized complex of cellulases according the initial shift of pH in the SSF.

The same hypothesis can be applied to *A. sydowii* CBMAI 934, which is better adapted to more acidic conditions (Fig. 2a). *P. citrinum* CBMAI 1186 and *A. sydowii* CBMAI 935 strains exhibited good adaptability over the range of pH values evaluated, proved by the low F-value, 24.3 and 5.6, respectively, despite of being

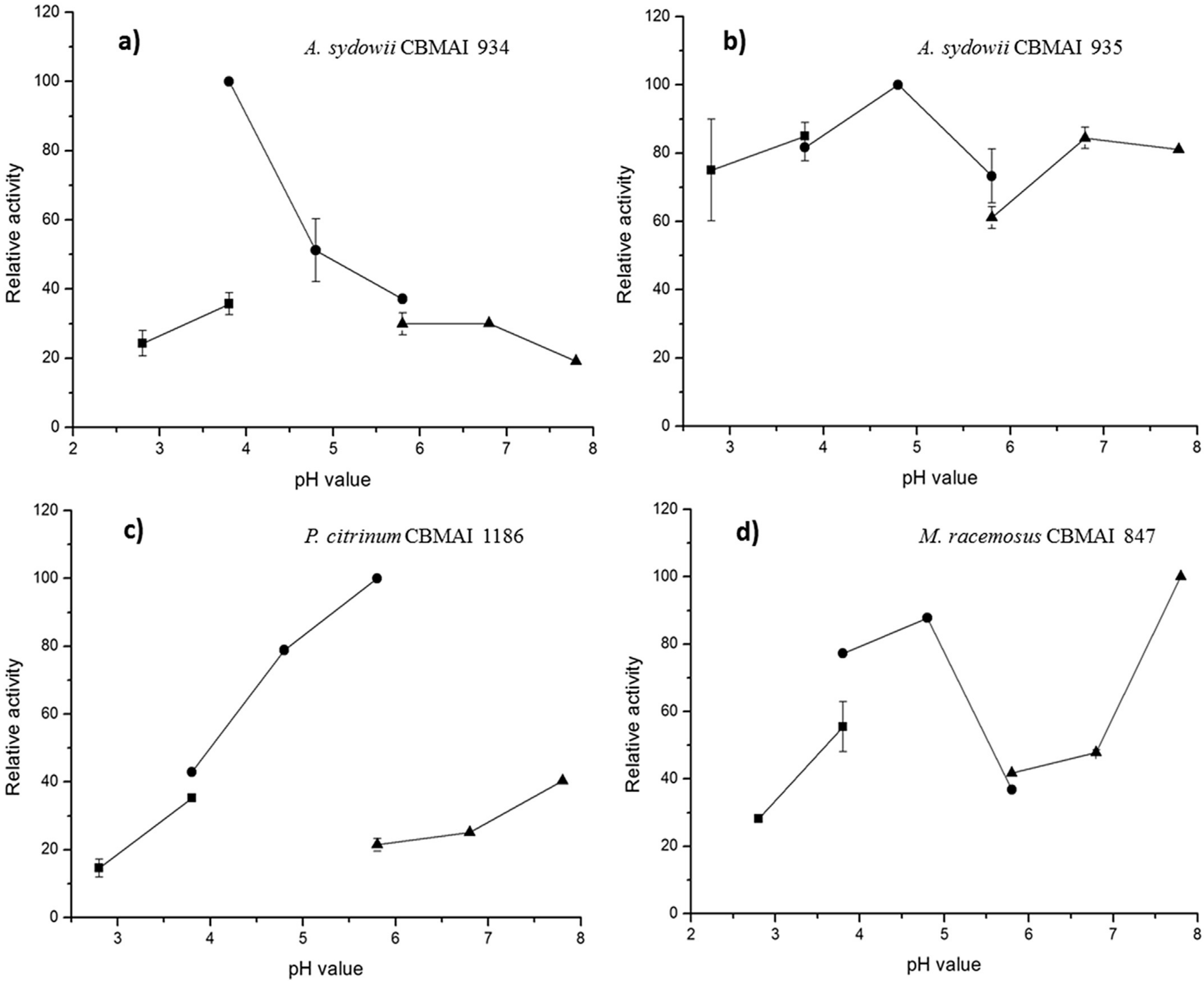


**Fig. 5.** Relative activity of crude extract obtained with different fungi strains by changing the pH value used in the filter paper hydrolysis[a]. [a]The activity 100% was the maximum FPase obtained for each value of pH tested during the hydrolysis of filter paper. Citric acid + $Na_2HPO_4$ buffer (—■—); citrate buffer (—●—); phosphate buffer (—▲—). The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (273,8), *A. sydowii* CBMAI 935 (17,9), *P. citrinum* CBMAI 1186 (4057,9) and *M. racemosus* (586,3).

statistically significant at 0.05 level (Fig. 2b and c). The adaptability of the fungus *A. sydowii* CBMAI 935 indicates that it might produce a balanced profile of different cellulases expressed by FPase, CMCase and β-glucosidase activities (Table 2), since there were not significant changes of activity for the different initial values of pH used in SSF. Trivedi et al. (2015) obtained good cellulase production by the marine-derived fungus *Cladosporium sphaerospermum*, cultivated in SSF, when the pH of the medium was adjusted to 4. Padmavathi et al. (2012) also studied the cellulase production by a marine-derived fungus strain, *Aspergillus* sp., and they obtained the highest activity when the fungus was cultivated in acidic conditions (pH 3). However, Sasi et al. (2012) reported that the marine-derived fungus strain *Aspergillus flavus* produced the highest activity when cultivated in pH 8. Therefore, these reports showed that marine-derived strains can biosynthesize cellulases in both acidic and alkaline conditions in SSF, accordingly the fungus strain, in agreement with the results of the present work.

For the temperature effect on cellulase biosynthesis, *M. racemosus* CBMAI 847 showed good cellulase production at two different temperatures, but it was best suited to 35 °C (Fig. 3d). *P. citrinum* CBMAI 1186 and *A. sydowii* CBMAI 935 also showed two optimal values, which reveals the abnormal adaptability of these fungi strains (Fig. 3b and c). The possible explanation for this behavior is similar to the presented previously in the pH study, i.e. changes in the availability of carbon sources that demand other cellulases or changes of amount of individual cellulases biosynthesized accordingly the temperature used in SSF. However, to the best of our knowledge, there were no previous studies of marine-derived fungi strains on the production of cellulases that corroborated this kind of behavior.

In the study of the effect of different volumes of Mandel solution, which is directly related to the moisture content in the solid medium, *P. citrinum* CBMAI 1186 and *A. sydowii* CBMAI 935 exhibited good adaptability over the volume range studied, as evidenced by the low F-values obtained from ANOVA (5,3 and 8,0, repectively) (Fig. 4). However, *M. racemosus* CBMAI 847 and *A. sydowii* CBMAI 934 demonstrated a maximum value of cellulase activity for one specific value of volume (12.5 and 7.5, respectively)

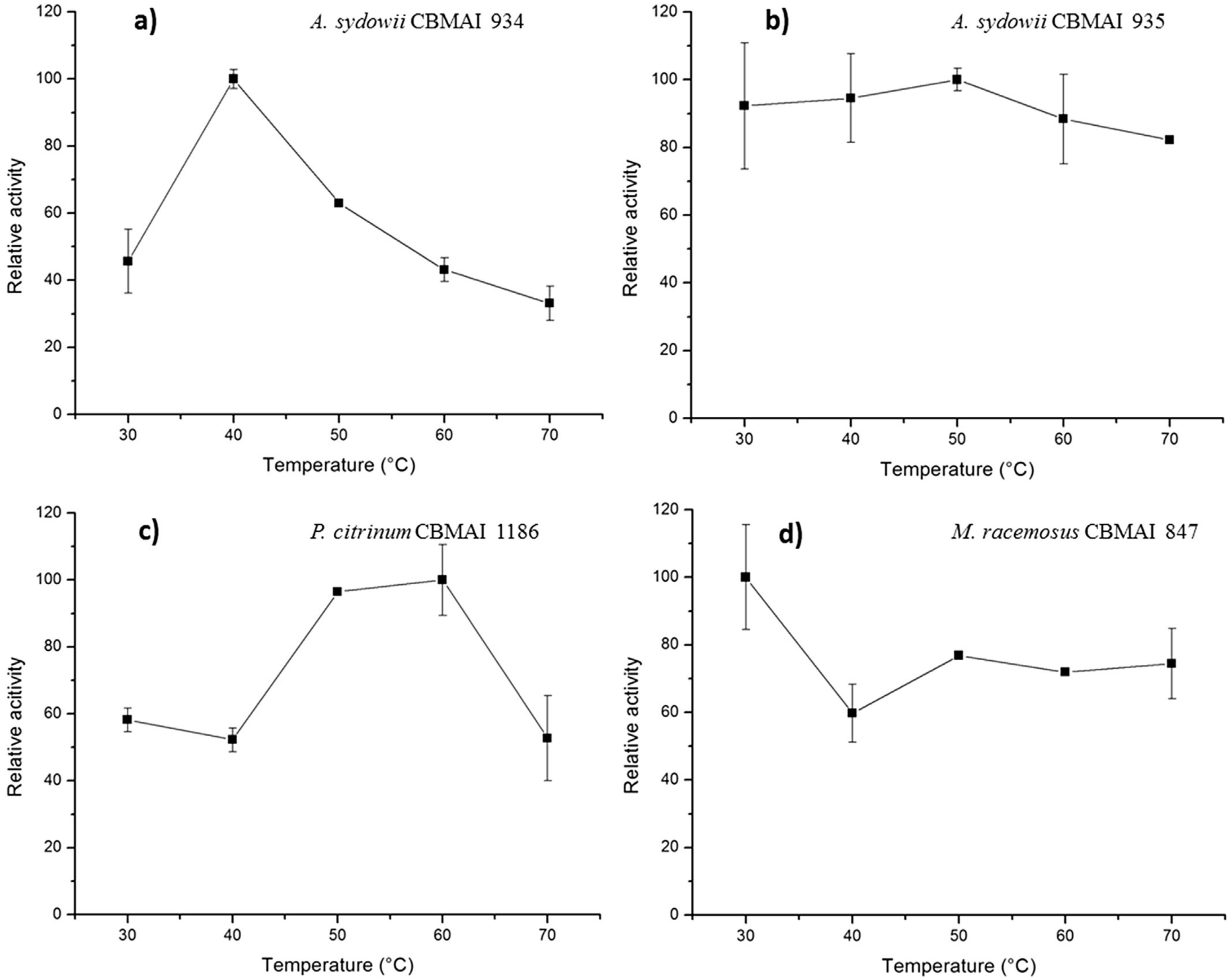


**Fig. 6.** Relative activity of crude extract obtained with different fungi strains by changing the temperature used in the filter paper hydrolysis reaction[a]. [a]The activity 100% was the maximum FPase obtained for each value of temperature tested during the hydrolysis of filter paper. The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (153,1), *A. sydowii* CBMAI 935 (1,9), *P. citrinum* CBMAI 1186 (59,1) and *M. racemosus* (15,2).

(Fig. 4). No morphological difference was observed in the fungi growth during the time that the strains were cultivated in different volumes of Mandel solution. Each strain has its own preferred moisture content and this is a crucial factor in SSF, once it influences the growth of the microorganisms and the secretion of metabolites, including the enzymes (Lee et al., 2011; Bansal et al., 2012; Raghuwanshi et al., 2014).

Based on these results, the *A. sydowii* CBMAI 935 strain exhibits good adaptability in all cultivation conditions studied (i.e., pH, temperature and volume of Mandel solution); this characteristic may be very important for industrial applications. *A. sydowii* CBMAI 934 was the only strain that exhibited one maximum cellulase activity for every variable. *P. citrinum* CBMAI 1186 and *M. racemosus* CBMAI 847 should be studied more carefully in order to determine their behavior during cultivation (mainly the influence of pH and temperature).

The influence of the parameters (pH, temperature and volume of crude extract) on the FPase assay was studied and different profiles and optimal conditions were obtained among the cellulases produced by the marine-derived fungi strains.

The cellulases produced by *A. sydowii* CBMAI 935 showed small changes in the activity along all the pH range studied but still has a maximum release of reducing sugars at pH 4.8 (Fig. 5b). The optimal activities obtained for *A. sydowii* CBMAI 934, *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186 cellulases are similar to those reported in the literature that ranged from 3.5 to 6 (Fadel, 2000; Ng et al., 2010; Liu et al., 2011) (Fig. 5). The activity of cellulases produced by *M. racemosus* CBMAI 847 showed that the highest FPase varied according to which buffer solution was used (pH 4.8 with a citrate buffer and pH 7.8 with a phosphate buffer) (Fig. 5d). This behavior for *M. racemosus* CBMAI 847 cellulases might occur due to the existence of two isoenzymes or subunits that have different optimal pH (4.8 or 7.8) for binding the polymer or catalytic action (Acharya et al., 2010). The cellulases produced by *M. racemosus* CBMAI 847 can be named as alkaline, once that one of its optimum pH was 7.8. This nomenclature was attributed accordingly the cellulases from *Bacillus carboniphilus* CAS 3 and *Bacillus licheniformis* AU01 with optimal pH values, pH 9 and 8–11, respectively (Annamalai et al., 2012, 2014).

The optimum temperature for the cellulases produced by

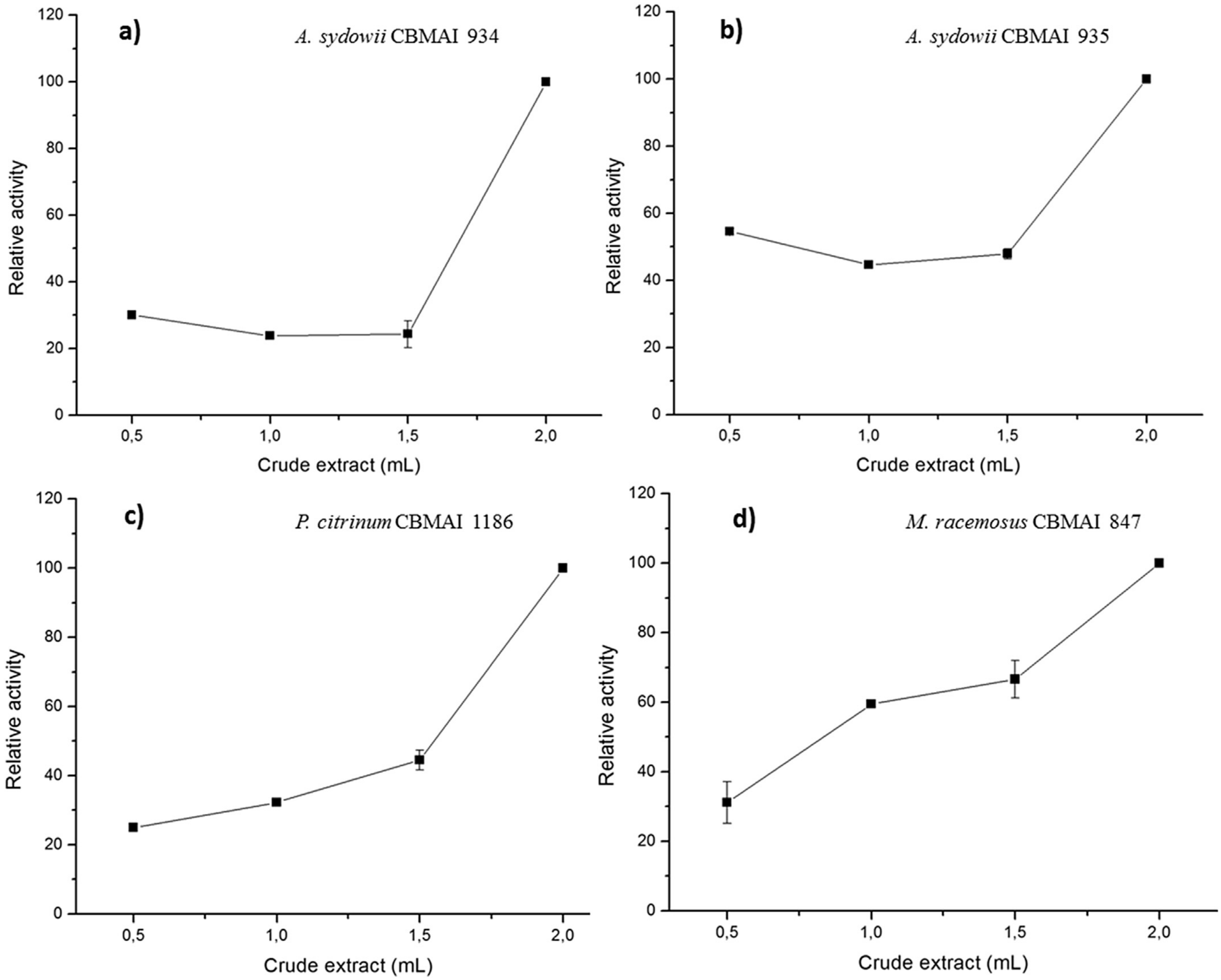


**Fig. 7.** Relative activity of crude extract obtained with different fungi strains by changing the enzyme volume used in the filter paper hydrolysis reaction[a]. [a]The activity 100% was the maximum concentration of reducing sugars obtained for each value of crude extract volume tested. The bars refers to the standard deviations. F-values from ANOVA: *A. sydowii* CBMAI 934 (2022,3), *A. sydowii* CBMAI 935 (4888,8), *P. citrinum* CBMAI 1186 (3497,6) and *M. racemosus* (297,6).

**Table 2**
FPase, CMCase and β-glucosidase activities after cellulase production improvement.

| Strain | FPase (FPU $g^{-1}$) | CMCase (U $g^{-1}$) | β-glucosidase (U $g^{-1}$) |
|---|---|---|---|
| *Aspergillus sydowii* CBMAI 934 | 1.92 ± 0.31 | 4.36 ± 0.08 | 0.14 ± 0.01 |
| *Aspergillus sydowii* CBMAI 935 | 0.73 ± 0.14 | 3.06 ± 0.03 | 0.72 ± 0.01 |
| *Penicillium citrinum* CBMAI 1186 | 1.33 ± 0.28 | 1.93 ± 0.04 | 0.22 ± 0.01 |
| *Mucor racemosus* CBMAI 847 | 2.41 ± 0.07 | 3.73 ± 0.03 | 0.52 ± 0.01 |

**Table 3**
Extraction of the adsorbed cellulases from wheat bran by a second washing step with acetate buffer.

| Strain | Residual FPase (%)[a] |
|---|---|
| *Aspergillus sydowii* CBMAI 934 | 60.3 ± 5.1 |
| *Aspergillus sydowii* CBMAI 935 | 37.1 ± 3.3 |
| *Penicillium citrinum* CBMAI 1186 | 34.6 ± 6.7 |
| *Mucor racemosus* CBMAI 847 | 25.1 ± 2.5 |

[a] Percentual relative activity by comparison with the first extraction activity.

*M. racemosus* CBMAI 847 was 30 °C much lower than the standard temperature determined by IUPAC and, to the best of our knowledge, there is no report of cellulases that have an optimum temperature of 30 °C (Fig. 6d) (Ghose, 1987). Because of this behavior, the use of the cellulases from *M. racemosus* CBMAI 847 in industrial processes would require low energy consumption during the hydrolysis of cellulose. The cellulases produced by *A. sydowii* CBMAI 935 maintained their maximum activity in the temperatures between 30 and 60 °C (Fig. 6b). Then, these cellulases would also require low energy consumption during the hydrolysis of cellulose but also can keep their activity at high temperatures if the

**Table 4**
Chemical composition and percentage of saccharification of integral sugarcane bagasse (SCB), untreated and treated, and its treated fractions (pith and rind) by crude cellulase extract from marine-derived fungi strains (*M. racemosus* CBMAI 847, *A. sydowii* CBMAI 934, *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186) cultivated under SSF.

| Bagasse sample | Cellulose (%) | Hemicellulose (%) | Lignin (%) | Saccharification (%) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 847[a] | 934[b] | 935[c] | 1186[d] |
| Untreated (SCB) | 42.7 | 26.0 | 26.5 | 20.0 | 21.0 | 0 | 0 |
| Treated (SCB) | 63.8 | 26.2 | 10.0 | 24.0 | 78.0 | 19.0 | 19.0 |
| Pith (treated) | 64.7 | 26.4 | 9.4 | 37.0 | 81.0 | 32.0 | 28.0 |
| Rind (treated) | 70.0 | 25.7 | 9.0 | 30.0 | 56.0 | 3.0 | 4.0 |

[a] *M. racemosus* CBMAI 847.
[b] *A. sydowii* CBMAI 934.
[c] *A. sydowii* CBMAI 935.
[d] *P. citrinum* CBMAI 1186.

industrial processes require that.

The variability in the best FPase assay parameters achieved for the different marine-derived fungi strains when using different temperatures and pH is also related to genetical diversity present in fungi from both land and water (i.e., adaptability to the environment) (Richards et al., 2011). The observed variations among these four marine-derived fungi strains imply that they could be specifically selected and used over a large range of pH and temperature values and can also be useful for specific cellulase application.

The volume of crude extract also influenced the release of sugars from the filter paper. However, this effect was observed only when the amount of cellulases was twice the standard value (2.0 mL); values of 0.5, 1.0 and 1.5 mL did not significantly affect the quantity of reducing sugars released according to the DNS method (Fig. 7).

After the optimization studies of fungi cultivation and the cellulase biosynthesis under SSF and the reducing sugars release of the filter paper, the final FPase obtained was significantly improved for all the fungi strains (Table 2). *A. sydowii* CBMAI 935 produced cellulases with the lowest FPase and *M. racemosus* CBMAI 847 exhibited the highest FPase (Table 2). However, in terms of the CMCase activity, the lowest value was found for *P. citrinum* CBMAI 1186, which was the strain that also exhibited the smallest improvement in FPase. The CMCase and β-glucosidase activity of *A. sydowii* CBMAI 935 was reasonably balanced when compared with the other strains. However, as mentioned above, *A. sydowii* CBMAI 935 has low FPase, which is probably related to the fact that this strain is not a good exo-cellulase producer, once it has produced endo-cellulase (CMCase) and β-glucosidase in reasonable amounts in comparison with the other marine-derived fungi strains. The lowest β-glucosidase activity (0.14 U $g^{-1}$) was produced by *A. sydowii* CBMAI 934, although this strain had the highest CMCase activity (4.36 U $g^{-1}$) (Table 2). Because of its low β-glucosidase activity, the FPase of this strain was lower than that observed for *M. racemosus* CBMAI 847 (Table 2).

This work proves that the catalytic action of cellulase complex is a contribution of several and different enzymes included into the endo- and exo-cellulases groups working concomitantly and synergistically with β-glucosidase.

The values of FPase (Table 2) were compared with terrestrial fungi also cultivated in wheat bran (Table 5).

Our results represent the first report of cellulase production by these marine-derived fungi strains, and they are lower than some results described for cellulase production by terrestrial strains. The best result obtained from a terrestrial strain was 10 times higher (*Trichoderma reesei*; 22.89 FPU $g^{-1}$) than that obtained from *M. racemosus* CBMAI 847 (2.4 FPU $g^{-1}$; Table 2) (Dhillon et al., 2011).

On the other hand, the study of marine-derived strains contributes to the knowledge about the possibility of using cellulases from marine environment in the saccharification of lignocellulosic raw materials.

In spite of the low activity (FPase) of cellulases produced by these marine-derived fungi in comparison with terrestrial strains (Table 5), as we are going to show forward, they are excellent catalytic tools on saccharification of the sugarcane bagasse.

When the cellulases produced in this work were applied to the cellulose hydrolysis from the sugarcane bagasse, it was observed that the percentage of saccharification was inversely proportional to the amount of lignin in the lignocellulosic raw material (Table 4). Thus, the removal of lignin from sugarcane bagasse resulted in an increase of the released reducing sugars, i.e. the saccharification promoted by the cellulases from *A. sydowii* CBMAI 934 shifted from 21% to 78%. These results corroborate with what is already well known that lignin imbedded in the polysaccharide matrix may block the access of cellulases to cellulose (Galbe and Zacchi, 2012). Moreover, Qin et al. (2014) noted that a commercial cellulase system from *Trichoderma reesei* exhibited a 45% stronger attachment adsorption to lignin than to hydroxypropyl cellulose and, as a consequence, a lower amount of “free” cellulase remained for the hydrolysis of cellulose.

After the pretreatment of sugarcane bagasse and its fractions with sodium hydroxide solution, the *A. sydowii* CBMAI 934 strain exhibited consistent amount of saccharification (56–81%, Table 4), in contrast with the saccharification performed by the cellulases produced by *M. racemosus* CBMAI 847, *A. sydowii* CBMAI 935 and *P. citrinum* CBMAI 1186. These results imply that the profile of individual enzymes in the cellulase complexes produced by these different fungi strains showed different potential for their application in the process of sugarcane bagasse saccharification even when using the same FPase value. The highest saccharification efficiency showed by the cellulases from *A. sydowii* CBMAI 934 could be associated to the highest CMCase activity in relation to the other marine-derived fungi strains (Table 2). The best performance of this strain could be a consequence of the high activity of endo-cellulases but probably there are others lignocellulolytic enzymes (such as hemicellulases and/or laccases) in the crude extract that contribute to the high saccharification. The method employed for sugar quantification (DNS) did not discriminate the source of reducing sugars as it may also be possible that different sugars (xylose, glucose, arabinose, cellobiose, etc.) were released by enzymes from *A. sydowii* CBMAI 934. This hypothesis was confirmed by the percentage of cellulose (63.8%) present in the dry mass of sugarcane bagasse after pretreatment when there was 78.0% of saccharification (Table 4).

Treatment of the pith and rind fractions with sodium hydroxide solution have almost identical chemical composition, but they exhibited different percentage of saccharification, i.e. pith achieved 81% (Table 4).

Sugarcane bagasse is mainly composed of two different types of cells. The rind contains mostly fibrous sclerenchyma cells that are responsible for the main structural support of sugarcane and have a

**Table 5**
FPU activities of terrestrial fungi from literature and marine-derived fungi strains using wheat bran as carbon source in SSF.

| Strain | FPase (FPU $g^{-1}$) | References |
|---|---|---|
| *Chrysoporthe cubensis* | 2.52 | Falkoski et al. (2013) |
| *Trichoderma reesei* RUT30 | 22.89 | Dhillon et al. (2011) |
| *Thermoascus auranticus* | 4.40 | Kalogeris et al. (2003) |
| *Fomitopsis* sp. RCK2010 | 3.49 | Deswal et al. (2011) |
| *Trichoderma asperellum* RCK2011 | 1.60 | Raghuwanshi et al. (2014) |
| *Aspergillus niger* MTCC 7956 | 4.55 | Sukumaram et al., 2009 |
| *Aspergillus niger* NS-2 | 16.8 | Bansal et al. (2012) |
| *P. janthinellum* NCIM 1171 mutant EU2D-21 | 10.0 | Singhvi et al. (2011) |
| *Aspergillus sydowii* CBMAI 934 | 1.90 | This work |
| *Aspergillus sydowii* CBMAI 935 | 0.73 | This work |
| *Penicillium citrinum* CBMAI 1186 | 1.30 | This work |
| *Mucor racemosus* CBMAI 847 | 2.40 | This work |

thick, lignified secondary wall. On the other hand, the pith portion, located in the inner part of the stem, contains mostly parenchyma cells that are roughly spherical, with thin and flexible cell walls, and is responsible for the storage of sucrose (Wirawan et al., 2010; Galbe and Zacchi, 2012).

Pith fractions demonstrated a high saccharification yield for all the studied fungi strains, when compared to rind and integral samples. This finding confirms that the inner part of sugarcane stalks is more susceptible to hydrolysis than the rind. For this reason, the integral sugarcane bagasse was less susceptible to hydrolysis than the pith and more susceptible to hydrolysis than the rind, as it is made up of both of these fractions. Higher susceptibility of pith cells to enzymatic hydrolysis was also observed by Mendes et al. (2016) in a study on the anatomic and ultrastructural characteristics of sugarcane.

Based on these results, it is possible to conclude that the four marine-derived strains studied in this work are good cellulase producers. Moreover, they might also be producers of hemicellulases and/or laccases, and they should be tested with other lignocellulosic raw materials. Additionally, the study of new pretreatment methods for total lignin removal is an important way of improving the hydrolysis of lignocellulosic material by the cellulases produced by these marine-derived fungi strains.

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